

$$\boxed{\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}(J)} \quad \text{General version}$$

(Fourier's law)

Linear Diffusion: $J = -k \frac{\partial u}{\partial z} \rightarrow \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial z^2}$

Nonlinear Diffusion: $J = -\frac{\alpha}{u^2} \frac{\partial u}{\partial z} \rightarrow \frac{\partial u}{\partial t} = \frac{\partial}{\partial z} \left(\frac{\alpha}{u^2} \frac{\partial u}{\partial z} \right)$

(cell movement)

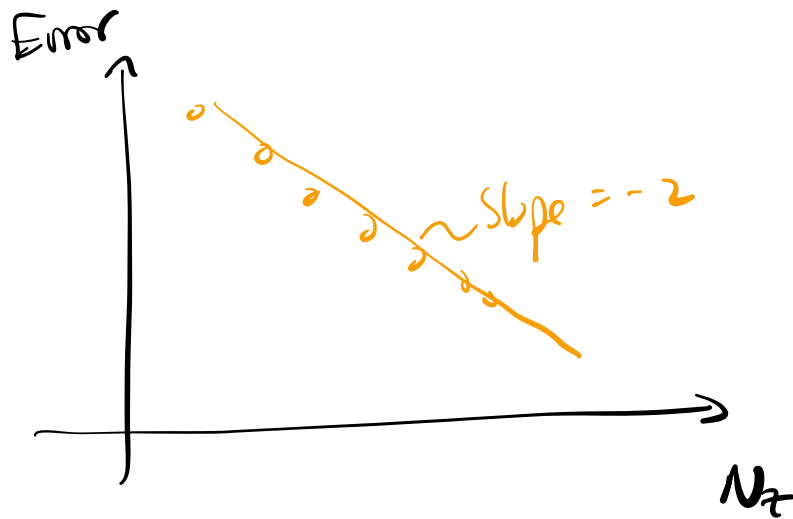
$$\frac{\partial u}{\partial t} = D_z * (\alpha / u \cdot 12 * (D_z * u))$$

What is D_z ? look up on Wikipedia
finite diff schemes.

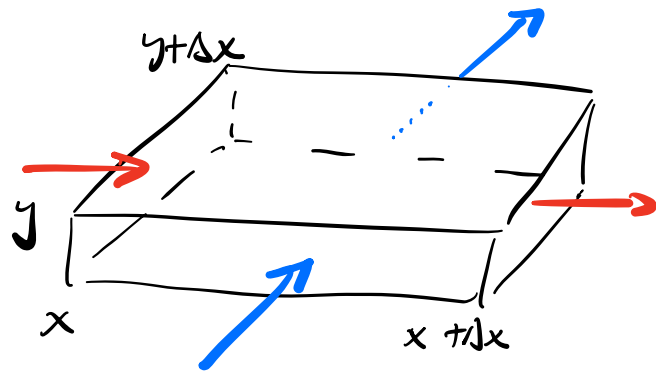
Linear
Convective velocity: $J = ku \rightarrow \frac{\partial u}{\partial t} = -k \frac{\partial u}{\partial z}$

↳ traffic flow

Nonlinear Convection: $J = f(u) \cdot u \rightarrow \frac{\partial u}{\partial t} = -\frac{\partial}{\partial z} (f(u) \cdot u)$



Lower
Diffusion in 2D:



$$\frac{\partial u}{\partial t} = -\frac{\partial}{\partial x}(J_x) - \frac{\partial}{\partial y}(J_y)$$

2D Heat: $J_x = -k_x \frac{\partial u}{\partial x}$
 $J_y = -k_y \frac{\partial u}{\partial y}$ $\rightarrow \frac{\partial u}{\partial t} = k_x \frac{\partial^2 u}{\partial x^2} + k_y \frac{\partial^2 u}{\partial y^2}$

3D Heat: $\frac{\partial u}{\partial t} = k_x \frac{\partial^2 u}{\partial x^2} + k_y \frac{\partial^2 u}{\partial y^2} + k_z \frac{\partial^2 u}{\partial z^2}$

$\frac{\partial u}{\partial t} = -\nabla \cdot \langle J_x, J_y \rangle$ Vector Field

$$\xrightarrow{\text{heat}} \frac{\partial u}{\partial t} = -\nabla \cdot \left\langle -k \frac{\partial u}{\partial x}, -k \frac{\partial u}{\partial y} \right\rangle$$

$$\frac{\partial u}{\partial t} = k \nabla \cdot \nabla u$$

$$\boxed{\frac{\partial u}{\partial t} = k \Delta u} = k \frac{\partial^2 u}{\partial x^2} + k \frac{\partial^2 u}{\partial y^2}$$

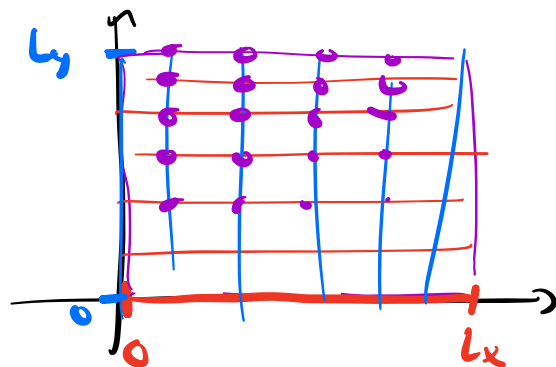
$$u = u(x, y, t)$$

$$\frac{\partial u}{\partial t} = k_x \frac{\partial^2 u}{\partial x^2} + k_y \frac{\partial^2 u}{\partial y^2}$$

$$x \in [0, l_x], y \in [0, l_y], t > 0$$

$$u(x, y, t) \approx u(x_i, y_j, t)$$

$$U = \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n_y} \\ u_{21} & u_{22} & \dots & u_{2n_y} \\ \vdots & \vdots & \ddots & \vdots \\ u_{n_x1} & \dots & \dots & u_{n_x n_y} \end{bmatrix}$$



$$\frac{\partial U}{\partial t} = k_x D_{xx} U + k_y U D_{yy}^T$$

$$\lim_{t \rightarrow \infty} u = \frac{\int_0^{L_x} \int_0^{L_y} u(x, y, t) dy dx}{\int_0^{L_x} \int_0^{L_y} dy dx}$$

for $t = 0$

